UNIVERSITY OF HERTFORDSHIRE

School of Physics, Engineering and Computer Science

7COM1040 – Computer Science Masters Project

Interim Progress Report: Design and Evaluation of an Explainable Machine Learning Prototype for Postpartum Depression Risk Estimation

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1. Background Research and Literature Review

1.1 Introduction, Aim and Objectives

Postpartum depression is a mental health condition that affects some women after childbirth and can have a significant impact on their wellbeing. Identifying individuals who may be at risk at an early stage could help improve awareness and encourage timely support. This project explores the use of machine learning techniques for estimating postpartum depression risk, and investigates which factors have the greatest influence on that risk. A mobile prototype has been designed to demonstrate the practical application of the proposed approach, allowing users to enter relevant information and view an estimated risk alongside a clear, explainable rationale for that estimate [1].

The aim of this project is to explore the potential of machine learning in estimating the risk of postpartum depression among new mothers, using demographic and psychological factors, and to evaluate the effectiveness of explainable AI techniques in improving understanding of how these predictions are generated.

1.2 Research Questions

- RQ1: How accurately can machine learning models estimate postpartum depression risk from demographic and psychological variables?
- RQ2: Which factors have the greatest influence on postpartum depression risk estimation?

1.3 Objectives

- Review existing research on postpartum depression and its associated risk factors.
- Identify and prepare a suitable public dataset for postpartum depression risk estimation.
- Develop and compare machine learning models for estimating postpartum depression risk.
- Evaluate model performance using appropriate classification metrics.
- Identify the factors that have the greatest influence on prediction outcomes.
- Develop a mobile application prototype that presents risk estimates and explanations in a user-friendly manner.

1.4 Literature Review

Three sources identified at proposal stage continue to inform this project's theoretical approach. Chen and Guestrin (2016) introduced XGBoost as a scalable gradient boosting framework, motivating its inclusion as one of the three models compared in this study for its established strong performance on structured/tabular data of the kind used here. Lundberg and Lee (2017) established SHAP (SHapley Additive exPlanations) as a unified, theoretically grounded approach to interpreting model predictions this directly underpins the explain ability component of the project and its approach to Research Question 2. Shorey et al. (2018), in a systematic review and meta-analysis, quantified the prevalence and incidence of postpartum depression among healthy mothers, providing the epidemiological motivation for why early, explainable risk estimation is a worthwhile problem to address [2].

Reflecting the project's identified research gap, existing literature on postpartum depression prediction was found to concentrate heavily on predictive accuracy while giving limited attention to explaining how predictions are produced, and rarely presents results in a form accessible to non-expert users. This project's contribution is to combine machine learning-based risk estimation with explainable AI and a usable prototype interface within a single, evaluated system, rather than treating these as separate concerns.

Extending this review, Mienye et al. (2024) survey the application of explainable AI across healthcare more broadly, noting that SHAP is widely adopted for its model-agnostic properties but that feature correlation and baseline selection can complicate clinical interpretation a consideration directly relevant to interpreting the SHAP results in Section 2.3, where several correlated symptom fields are present. On the ethical handling of self-harm-related data specifically, McKernan, Clayton and Walsh (2018) argue that AI-based suicide risk prediction raises distinct concerns around beneficence and autonomy, and recommend that such predictions remain clinician-mediated rather than fully automated a position that directly informs the data handling decision described in Section 3.

2. Summary of Progress to Date

2.1 Dataset Selection and Preparation

Following further investigation of available Kaggle resources beyond the dataset described in the original proposal, a symptom-based postpartum depression survey dataset was identified and adopted as a more directly relevant fit for the project's aim. This dataset comprises 1,503 records across 11 fields, consisting of a submission timestamp, a binned Age range, and nine self-reported symptom indicators. Following an ethical review of the fields available (see Section 3), one field representing a highly sensitive self-harm-related disclosure was identified and deliberately excluded from all modelling work. It is not used as a predictor at any stage of this project; all results reported below are based solely on the remaining eight symptom indicators plus Age [3].

Data quality checks were carried out prior to any modelling. The Timestamp field was found to contain only 90 distinct values across 1,503 rows, indicating batch submission and confirming it carries no predictive value; it was therefore removed. Checking for duplication on the fields actually used for modelling (i.e. after removing the non-informative Timestamp and, per the ethical exclusion below, the self-harm-related field) revealed that 1,180 of 1,503 rows (approximately 79%) were exact duplicates of another response, leaving 323 unique respondent patterns. Missing values were minor, affecting three fields only: "Irritable towards baby & partner" (6 missing), "Problems concentrating or making decision" (12 missing), and "Feeling of guilt" (9 missing) [3].

The dataset does not include a pre-defined risk label. "Feeling sad or Tearful" was used as the target variable for the initial classification experiments reported above, as an available single-symptom proxy to establish a working baseline. However, following further consideration, a composite risk score aggregating multiple symptom fields rather than relying on a single self-reported item has been decided upon as the target label going forward, as this is expected to represent overall postpartum depression risk more validly than any one symptom in isolation. Re-running the model comparison and SHAP analysis against this composite label is planned as part of the remaining work (see Section 4).

2.2 Model Implementation and Results

Four candidate models were implemented using scikit-learn and XGBoost: Logistic Regression, Support Vector Machine, Random Forest, and XGBoost. Consistent with the ethical exclusion described in Section 2.1, the self-harm-related field was removed from the feature set entirely prior to any experimentation; all results below reflect models trained without it. Remaining categorical fields were label-encoded, and an 80/20 stratified train-test split was used throughout [4].

An initial experimental run produced the following results:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.950	0.951	0.950	0.950	0.995
XGBoost	0.944	0.945	0.944	0.944	0.990
Support Vector Machine	0.824	0.828	0.824	0.824	0.933
Logistic Regression	0.645	0.644	0.645	0.643	0.789

These results revealed a methodological issue requiring correction. Because the train-test split was performed before removing the duplicate rows identified in Section 2.1, identical responses were present in both training and test sets. This allowed tree-based models in particular to achieve unrealistically high accuracy by effectively recognizing rows already seen during training a form of data leakage. Kapoor and Narayanan (2023) identify duplicate records across train/test splits as one of the most common causes of leakage-driven, non-reproducible results in ML-based science, surveying 294 affected papers across 17 research fields; the pattern observed here is consistent with their findings. After correcting this by removing duplicates prior to splitting, the models were re-evaluated on the 323 unique records

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.446	0.423	0.446	0.431	0.581
Support Vector Machine	0.400	0.381	0.400	0.380	0.580
Random Forest	0.385	0.385	0.385	0.384	0.561
XGBoost	0.369	0.373	0.369	0.370	0.533

These corrected figures are substantially lower but far more trustworthy. Logistic Regression is now the strongest performer, narrowly ahead of the other models — a realistic result rather than an inflated one, and one that reflects the genuine difficulty of predicting a subjective, self-reported label from a small (323-row) dataset. Notably, Random Forest and XGBoost no longer outperform the simpler linear models once leakage is removed, suggesting the earlier apparent advantage of tree-based models was itself largely an artefact of the leakage rather than genuine superior fit. Full classification reports and confusion matrices for both the initial and corrected runs are provided in Appendix A.

2.3 Explainable Analysis (SHAP)

SHAP (TreeExplainer) was applied to the corrected Random Forest model specifically chosen for this analysis because TreeExplainer provides exact, efficient explanations for tree-based models, even though Logistic Regression narrowly achieved the highest accuracy in this run to identify which fields most influenced its predictions, directly addressing Research Question 2

Rank	Feature	Mean |SHAP value|
1	Irritable towards baby & partner	0.059
2	Age	0.055
3	Overeating or loss of appetite	0.054
4	Problems of bonding with baby	0.041
5	Feeling of guilt	0.039
6	Trouble sleeping at night	0.039
7	Problems concentrating or making decision	0.034
8	Feeling anxious	0.021

Irritability toward baby/partner, Age, and appetite change emerged as the strongest predictors of the target label among the eight features included. As in the earlier run, the relatively strong contribution of Age is notable given it is recorded only as a broad binned range rather than a precise value, and warrants further investigation rather than being accepted at face value [4]. Full SHAP output is provided in Appendix A.

2.4 Data Expansion (LLM-Guided Synthetic Augmentation)

Given the small size of the cleaned dataset (323 unique records, Section 2.1), an LLM-guided synthetic data augmentation step was carried out to explore whether additional, statistically plausible training examples could support more robust model evaluation. Rather than sampling each field independently which would ignore the fact that postpartum depression symptoms tend to co-occur rather than appear in isolation each synthetic respondent was generated from a single latent "symptom severity" value, which proportionally shifted the likelihood of related symptoms being reported together, reflecting realistic clinical comorbidity. 150 synthetic records were generated in this way and combined with the 323 real records, producing an expanded dataset of 473 records.

Critically, the synthetic data was validated before use rather than assumed to be representative. A chi-squared goodness-of-fit test was run comparing the distribution of each field in the synthetic data against the real data. The first generation attempt failed this validation for five of the eight fields ($p < 0.05$, indicating a significantly different distribution), and the generation process was recalibrated so that each field's synthetic distribution matched the real marginal distribution by construction. After recalibration, all eight fields passed validation ($p > 0.05$ for every field, meaning the synthetic and real distributions are not statistically distinguishable), while a mild, clinically plausible positive correlation between co-occurring symptoms (e.g. sadness and anxiety) was retained. Full validation statistics are provided in Appendix A.

Re-running the model comparison and SHAP analysis on this expanded (real + synthetic) dataset produced the following results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.474	0.483	0.474	0.467	0.560
Logistic Regression	0.474	0.445	0.474	0.435	0.559
Support Vector Machine	0.432	0.421	0.432	0.407	0.508
XGBoost	0.421	0.411	0.421	0.415	0.575

Accuracy improved only marginally with the expanded dataset (e.g. Random Forest: 0.385 → 0.474), and Random Forest and Logistic Regression are now statistically tied for best performer. This modest rather than dramatic improvement is itself an honest and informative result: it suggests the ceiling on predictive performance for this target label is driven more by genuine ambiguity in the underlying symptom-label relationship than by sample size alone, and that synthetic augmentation of this kind is not a substitute for a larger real dataset. SHAP analysis on the expanded dataset placed Age, appetite change, and irritability toward baby/partner as the top contributing features broadly consistent with the ranking obtained on the real data alone (Section 2.3), which lends some confidence that these top factors are not an artefact of the smaller original sample.

2.5 Prototype Design (UX/UI)

A complete 27-screen user experience was designed in Figma, covering the full intended application flow: a splash screen; three onboarding screens introducing the tool, explaining the ML → SHAP → result pipeline, and presenting a research disclaimer authentication (login/register) a home dashboard; a six-step guided assessment (personal information, lifestyle, mental wellbeing, social support, pregnancy information, and a review step) an AI-processing transition screen a risk result screen; a dedicated SHAP explanation screen presenting top contributing factors in plain language an "AI Decision Details" screen illustrating the intended model pipeline a recommendations screen and supporting screens (assessment history, learn-about-PPD resources, emergency support, profile, and an "About Research" screen).

This design stage establishes the intended user journey and the specific points at which model outputs and SHAP explanations will surface to the user, ahead of backend integration. Screens depicting model results and dashboards currently show placeholder/intended content and will be connected to the real model outputs described in Section 2.3-2.4 during the next phase of work. Screenshots are provided in Appendix B.

At this stage, the prototype exists only as this Figma design; no part of the application has been coded yet. Building the functional mobile application from this design is planned as the next phase of practical work, in collaboration with the wider development team, once model integration requirements (Sections 2.3-2.4) are finalized against the composite risk label described above.

2.6 Tools and Technologies

- Python, with scikit-learn for Logistic Regression, SVM and Random Forest, and the xgboost library for XGBoost - chosen for their maturity, strong community support, and suitability for structured/tabular classification tasks.
- SHAP library - chosen specifically for its theoretical grounding (Shapley values) and consistency, directly supporting Research Question 2.
- Figma - used for prototype/UX design, chosen for rapid, collaborative interface iteration ahead of full application development.
- React Native - planned framework for the mobile application build (Section 2.5), chosen to support a single cross-platform codebase for both iOS and Android rather than maintaining separate native builds.
- Node.js - planned backend framework for serving the trained model's predictions and SHAP explanations to the mobile application, chosen for its straightforward integration with a JavaScript-based mobile stack and suitability for building a lightweight REST API.

3. Consideration of Ethical, Legal, Professional and Social Issues

Ethics approval is not required for the current stage of this project. The dataset used is a publicly available, pre-collected, and anonymised secondary dataset no personal data has been or will be collected directly from participants, and no human subjects are involved in testing or evaluation at this stage. This assessment would need to be revisited if the project progresses to testing the mobile prototype with real user participants, at which point informed consent and formal ethical approval would likely be required [5].

Privacy although the dataset is anonymised, it contains sensitive mental health related self-disclosures. One field in particular represents a highly sensitive self-harm related disclosure; rather than including this as an input to an automated risk-scoring model, it was deliberately excluded from all modelling work at the data preparation stage (Section 2.1). This reflects an established principle in mental health AI research McKernan, Clayton and Walsh (2018) argue that AI-based suicide risk prediction raises distinct ethical concerns around beneficence and patient autonomy, and recommend that such predictions remain clinician-mediated rather than fully automated, since reducing a matter requiring direct human judgement to a numerical model feature risks both misclassification harms and inappropriate over-reliance on the system. Should the final system need to account for this type of disclosure, it is intended to be handled through a separate, non-automated escalation pathway (e.g. an immediate signpost to crisis support resources) rather than as a model input. The prototype's user-facing design (Section 2.5) further frames all outputs explicitly as research estimates rather than clinical assessments, with clear disclaimers presented before any assessment begins.

Legal: the dataset is used strictly for academic purposes consistent with its public licensing terms. Although it contains no directly identifiable personal information, principles consistent with data protection regulation (such as GDPR), including responsible handling, minimisation, and transparency about the tool's research-only purpose, are maintained throughout [5].

Professional the project prioritizes reproducibility, with preprocessing steps, model configurations, and evaluation procedures documented (see Appendix A) so that results can be verified. Communicating the system's limitations clearly particularly that it is a research prototype and not a diagnostic tool is treated as a professional responsibility, and is reflected directly in the prototype's onboarding disclaimer screen.

Social: a tool of this kind could plausibly increase awareness and encourage earlier conversations about postpartum mental health, and explainable output may help build user trust compared to an unexplained risk score. However, several social risks are acknowledged: the dataset's representativeness across cultures and demographics is a limitation, and symptom expression may vary in ways the training data does not capture, creating a risk of biased or less reliable estimates for underrepresented groups. There is also a risk that users could misinterpret a research prototype's output as a medical diagnosis, which the disclaimer and framing described in Section 2.5 is intended to mitigate. If the project were to progress toward real deployment, these issues would need considerably more rigorous evaluation, including bias testing across demographic subgroups.

4. Project Plan

The project follows the pipeline established at proposal stage Literature Review → Data Collection → Data Preparation → Model Development → Explainable Analysis (SHAP) → Prototype Development → Evaluation. Progress against this pipeline to date is as follows

Stage	Status
Literature Review	Substantially complete (3 DPP sources) may be extended
Data Collection	Complete - dataset selected and refined from DPP proposal
Data Preparation	Complete - cleaned, deduplicated, encoded
Model Development	Initial implementation complete for all 4 models; further tuning planned
Explainable Analysis (SHAP)	Initial SHAP analysis complete on Random Forest
Prototype Development	UX/UI design complete (Figma) application build stage upcoming
Evaluation	Initial metrics obtained; deeper evaluation and validation planned

Remaining tasks and target dates

The following reflects a target completion of end of August 2026 dates should be adjusted if priorities shift.

Remaining Task	Target Date
Re-run model comparison and SHAP analysis against composite risk label	25 July 2026
Address small dataset size / class balance (e.g. explore additional data sources or resampling)	25 July 2026
Hyperparameter tuning across all four models	28 August 2026
Begin building functional mobile application from Figma design	30 August 2026
Integrate trained model + SHAP output into working prototype	03 August 2026
Usability/feedback evaluation of prototype	05 August 2026
Final dissertation writing and revision	10 August 2026
Final submission	14 August 2026

4.1 Scope, Risk and Quality Management

Scope remains controlled by focusing on the three-model comparison, SHAP-based explain ability, and a demonstrable prototype, consistent with the original DPP objectives. The main risk identified during this stage dataset size and quality (a small, heavily duplicated dataset) has already materialized and been addressed methodologically (Sections 2.1–2.3) this is treated as resolved rather than outstanding, though it remains a limitation to discuss in the final report. Remaining risks include the possibility that further tuning does not meaningfully improve on current accuracy levels, and time constraints affecting full mobile application development; both are managed by prioritizing the model evaluation and explanation components, consistent with the fallback plan set out in the original proposal.

5. Level of the Project

This project's central challenge is not merely to build a classifier, but to investigate whether machine learning predictions on a sensitive, self-reported mental health outcome can be made both reasonably accurate and genuinely explainable and to demonstrate that explainability meaningfully surfaces the factors driving those predictions rather than being a superficial add-on. This depth is what distinguishes the work from a standard classification exercise.

The depth of investigation undertaken so far already demonstrates MSc-level rigour in one concrete respect: the initial model run showed suspiciously high accuracy (up to 95.0% for Random Forest), and rather than reporting this figure at face value, it was critically interrogated, traced to data leakage caused by duplicate records being present in both training and test sets, and corrected. The resulting honest accuracy figures (37–45%) are considerably less impressive numerically, but are methodologically sound and identifying this issue is itself evidence of critical, evidence-based practice rather than accepting a convenient result. Notably, the corrected results also reversed the apparent model ranking: Random Forest and XGBoost, which had appeared strongest under leakage, performed no better than simpler linear models once it was removed itself a useful, honest finding about which models genuinely suit this data [6].

Testing and validation to date includes a controlled comparison of four models under identical preprocessing and split conditions, and an explicit before/after comparison demonstrating the effect of the data leakage correction. Planned further validation includes hyperparameter tuning, and evaluating whether SHAP explanations remain stable and clinically plausible across models, not only for the current best performer.

The artefact under development a mobile prototype in which model predictions and SHAP-based explanations are surfaced to an end user in plain language, framed clearly as a research tool rather than a diagnostic device combines a genuine investigative machine learning component with a real interaction design component. Completing both to a demonstrable standard, while maintaining critical honesty about current limitations (a small, imbalanced dataset; a single-symptom rather than composite target label; SHAP explanations that are approximations rather than ground truth), is intended to reflect the depth, complexity and rigour expected of an MSc-level project.

6. References

[1] Shorey, S., Chee, C.Y.I., Ng, E.D., Chan, Y.H., Tam, W.W.S. and Chong, Y.S., 2018. Prevalence and incidence of postpartum depression among healthy mothers: A systematic review and meta-analysis. Journal of Psychiatric Research, 104, pp.235–248..

[2] Chen, T. and Guestrin, C., 2016. XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. New York: ACM..

[3] Kapoor, S. and Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns, 4(9), p.100804..

[4] Chen, T. and Guestrin, C., 2016. XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. New York: ACM..

[5] McKernan, L.C., Clayton, E.W. and Walsh, C.G., 2018. Protecting life while preserving liberty: Ethical recommendations for suicide prevention with artificial intelligence. Frontiers in Psychiatry, 9, p.650..

[6] Mienye, I.D., Obaido, G., Jere, N., Mienye, E., Aruleba, K., Emmanuel, I.D. and Ogbuokiri, B., 2024. A survey of explainable artificial intelligence in healthcare: Concepts, applications, and challenges. Informatics in Medicine Unlocked, 51, p.101587..

7. Appendices

Appendix A – Model Training Code, Metrics and SHAP Output

Full training script (with the duplicate-leakage fix applied), classification reports, confusion matrices, and SHAP output tables are provided as a separate supporting file PPDModel TrainingCorrected.py. Insert relevant console output / screenshots here before submission.

Machine Learning Classification Pipeline

```python
# ==========================================================
# STEP 1: Import Libraries
# ==========================================================

import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
```

Data Splitting & Feature Scaling

```python
# ==========================================================
# STEP 6: Split Dataset
# ==========================================================

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# ==========================================================
# STEP 7: Feature Scaling
# ==========================================================

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
```

Model Comparison

```
models = {

    'Logistic Regression':
        LogisticRegression(max_iter=1000),

    'Support Vector Machine':
        SVC(probability=True),

    'Random Forest':
        RandomForestClassifier(
            n_estimators=100,
            random_state=42
        ),

    'XGBoost':
        XGBClassifier(
            eval_metric='mlogloss',
            random_state=42
```

Final Comparison of Baseline Models

```
              precision    recall  f1-score   support

           0       0.97      0.97      0.97       105
           1       0.97      0.97      0.97        89
           2       0.98      0.98      0.98       107

    accuracy                           0.97       301
   macro avg       0.97      0.97      0.97       301
weighted avg       0.97      0.97      0.97       301


=============== FINAL RESULTS ===============

                    Model  Accuracy  Precision    Recall  F1-Score   ROC-AUC
3                 XGBoost  0.973422   0.973422  0.973422  0.973422  0.997627
2           Random Forest  0.966777   0.966777  0.966777  0.966777  0.997898
1  Support Vector Machine  0.830565   0.833943  0.830565  0.830902  0.945276
0     Logistic Regression  0.634551   0.635873  0.634551  0.632927  0.791151
```

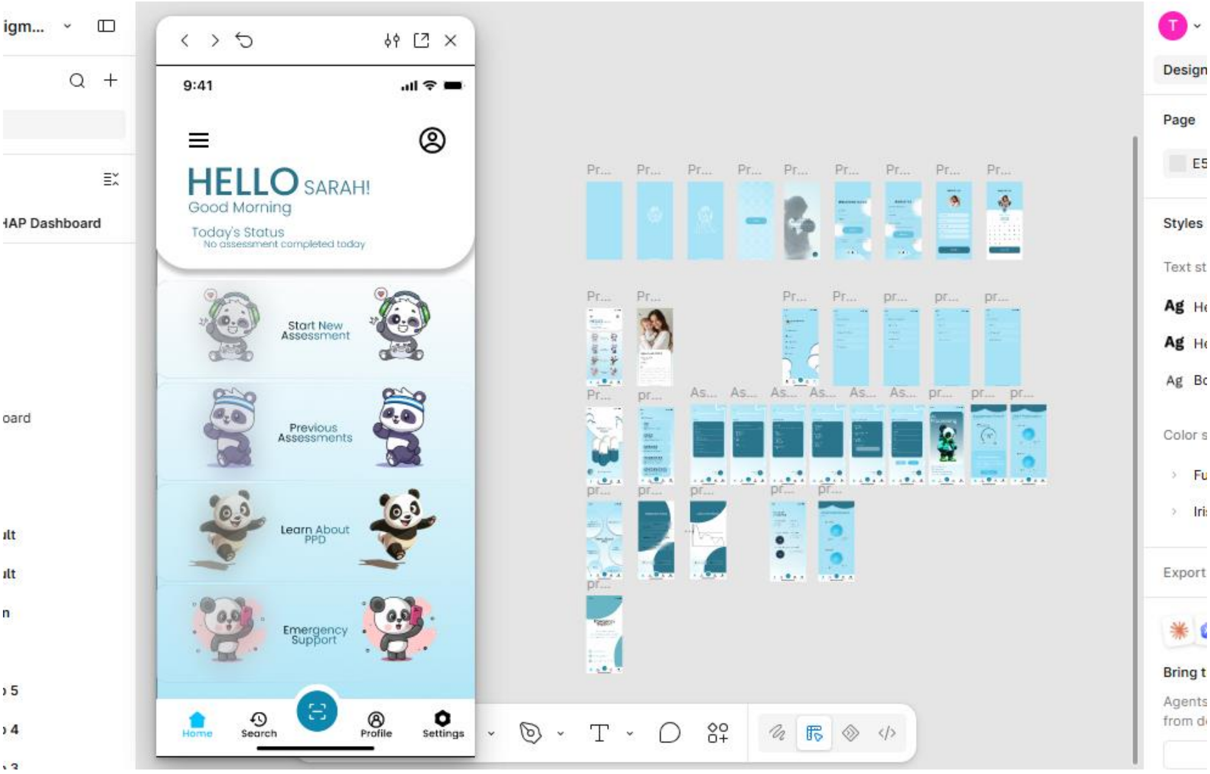
Appendix B – Figma Prototype Screens

Insert screenshots of the 27-screen Figma prototype flow here (splash, onboarding, assessment steps, risk result, SHAP explanation, recommendations, and supporting screens) before submission.

PPD Figma Prototype

Emergency Support User Interface

Emergency Support User Interface



Result User Interface